

Royston Soans

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SUMMARY

Full-stack developer specializing in efficient web applications, built with React, TypeScript, and Supabase. Shipped production-ready tools integrating LLM APIs with real-time backends, accessible UI systems, and gamified learning platforms.

EDUCATION

2024 – Present	Dwarkadas J. Sanghvi College of Engineering , Mumbai B.Tech in Computer Engineering SGPA: 9.64 (till 4th Sem)
2022 – 2024	MJ Junior College of Science, Mumbai HSC: 89.33%
2013 – 2022	Saint Francis School, Mumbai ICSE: 95%

PROJECTS

Poke-Learn

React, TypeScript, Supabase, TailwindCSS, Gemini API, Vite

- Architected a gamified learning platform across 5 themed regions with subject-based quiz progression and long-term user retention mechanics (badges, levels, streaks).
- Integrated Google Gemini API with prompt engineering to dynamically generate adaptive difficulty quiz questions, reducing repeated question exposure
- Implemented JWT-based authentication, persistent progress tracking, and real-time leaderboard ranking using Supabase and PostgreSQL.
- Designed pixel-art UI component system with TailwindCSS; shipped full-stack application solo from schema design to deployment.

ChromaGen

React, TypeScript, TailwindCSS, Groq API, Vite

- Built an AI-powered color palette generator supporting natural language prompts and image uploads, using LLM-based prompt parsing and palette ranking logic.
- Implemented WCAG AA/AAA accessibility validation for contrast ratios and readability scoring, enabling designers to surface compliance issues instantly.
- Engineered one-click CSS and design token export, reducing designer-to-developer handoff time.

YT-Audio Extractor

Node.js, Express.js, FFmpeg, yt-dlp, SSE

- Engineered a real-time YouTube audio streaming platform using Node.js, Express.js, yt-dlp, and FFmpeg, eliminating temporary file storage through a Unix pipe-based streaming architecture.
- Implemented dynamic audio processing with support for multiple formats (MP3, M4A, FLAC, OGG, WAV), including trimming, loudness normalization, metadata embedding, and album art injection.
- Developed live download progress tracking using Server-Sent Events (SSE), real-time metadata extraction for videos/playlists,

Packet Runner

Phaser 3, JavaScript, Vite

- Developed an interactive browser-based network routing puzzle game inspired by real networking concepts (VPN tunnels, firewalls, packet drop mechanics).
- Designed modular scene architecture and game loop in Phaser 3, with a custom scoring and combo system and progressive difficulty scaling across multiple levels.
- Implemented stateful node interaction system managing real-time packet routing logic.

SKILLS

Languages	C, C++, Java, Python, JavaScript, TypeScript
Frontend	HTML, CSS, React, TailwindCSS, Vite
Backend	Node.js, Supabase, PostgreSQL, REST API Design
AI / LLM	Gemini API, LLM Integration, Prompt Engineering, Adaptive AI Systems
Tools	Git, GitHub, WCAG Accessibility Standards